

Exposure instruments

BMI: 501 SNPs

Smoking initiation: 93 SNPs

Primary registry outcomes

FinnGen R5 IVDD: 20,001 cases

Spinal stenosis: 9,169 cases

164,682 controls for each endpoint

Primary univariable analyses

480 BMI / 85 smoking SNPs per outcome

IVW; four tests, $\alpha = 0.0125$

Weighted median; MR-Egger; diagnostics

Exploratory multivariable analyses

594 union SNPs; 1 coordinate exclusion

501 joint-LD-selected candidates

429 exact, nonpalindromic SNPs per outcome

Interpretation boundaries

Secondary: symptom endpoints, alternative adiposity, reverse MR, CAD comparison

No independent outcome replication; no measured zero-overlap claim

MRlap corrections and unverified UKB pain ORs excluded from current inference